

Algebra II with Trigonometry

Chapter 7.1

Olympiad Solutions (Gemini)

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Solutions

1. Evaluate the exact value of the following sum:

$$\log_2(\tan 1^\circ) + \log_2(\tan 2^\circ) + \log_2(\tan 3^\circ) + \cdots + \log_2(\tan 89^\circ)$$

Using the logarithmic identity $\log_a(x) + \log_a(y) = \log_a(xy)$, we can combine the sum into a single logarithm of a product:

$$\log_2(\tan 1^\circ \cdot \tan 2^\circ \cdot \tan 3^\circ \cdots \tan 89^\circ)$$

Recall the complementary angle identity $\tan(90^\circ - \theta) = \cot \theta$. We can pair each $\tan \theta$ from the first half of the product with $\tan(90^\circ - \theta)$ from the second half:

$$\tan 1^\circ \cdot \tan 89^\circ = \tan 1^\circ \cdot \cot 1^\circ = 1$$

$$\tan 2^\circ \cdot \tan 88^\circ = \tan 2^\circ \cdot \cot 2^\circ = 1$$

This pairing continues for all terms up to $\tan 44^\circ \cdot \tan 46^\circ = 1$. The only unpaired term in the exact middle is $\tan 45^\circ$, which is also equal to 1. Thus, the entire product evaluates to 1, leaving us with:

$$\log_2(1) = 0$$

2. Suppose x is an angle in the interval $(0, \pi)$ such that $\csc x - \cot x = 5$. Determine the exact value of $\cos x$.

We leverage the Pythagorean identity $\csc^2 x - \cot^2 x = 1$. Factoring this as a difference of squares yields:

$$(\csc x - \cot x)(\csc x + \cot x) = 1$$

Given that $\csc x - \cot x = 5$, we immediately find:

$$\csc x + \cot x = \frac{1}{5}$$

We now have a simple linear system in terms of $\csc x$ and $\cot x$. Subtracting the given equation from our new equation gives:

$$2 \cot x = \frac{1}{5} - 5 = -\frac{24}{5} \implies \cot x = -\frac{12}{5}$$

Adding the two equations yields:

$$2 \csc x = 5 + \frac{1}{5} = \frac{26}{5} \implies \csc x = \frac{13}{5} \implies \sin x = \frac{5}{13}$$

Finally, using the definition $\cot x = \frac{\cos x}{\sin x}$, we multiply to find $\cos x$:

$$\cos x = \cot x \cdot \sin x = \left(-\frac{12}{5}\right) \left(\frac{5}{13}\right) = -\frac{12}{13}$$

(Note: $\cos x < 0$ and $\sin x > 0$ correctly places x in Quadrant II, consistent with $x \in (0, \pi)$.)

3. Find the exact numerical value of the following trigonometric expression: $\sin^6(2024^\circ) + \cos^6(2024^\circ) + 3 \sin^2(2024^\circ) \cos^2(2024^\circ)$

Let $\theta = 2024^\circ$. Consider the algebraic identity

$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$. Let $a = \sin^2 \theta$ and $b = \cos^2 \theta$. By the fundamental Pythagorean identity, we know $a + b = \sin^2 \theta + \cos^2 \theta = 1$.

Substitute a and b into the algebraic identity:

$$\begin{aligned} (\sin^2 \theta + \cos^2 \theta)^3 &= (\sin^2 \theta)^3 + (\cos^2 \theta)^3 + 3(\sin^2 \theta)(\cos^2 \theta)(\sin^2 \theta + \cos^2 \theta) \\ 1^3 &= \sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta(1) \end{aligned}$$

Therefore, the expression perfectly collapses, completely independent of the angle, to:

4. By considering an appropriate trigonometric substitution, determine the maximum possible value of the algebraic function $f(x) = \frac{x}{x^2+25}$ over all real numbers x .

To simplify the denominator, let $x = 5 \tan \theta$, where $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Substituting this into $f(x)$ gives:

$$f(5 \tan \theta) = \frac{5 \tan \theta}{25 \tan^2 \theta + 25}$$

Factor out the 25 in the denominator and apply the identity $\tan^2 \theta + 1 = \sec^2 \theta$:

$$f(5 \tan \theta) = \frac{5 \tan \theta}{25 \sec^2 \theta} = \frac{1}{5} \left(\frac{\sin \theta}{\cos \theta} \cdot \cos^2 \theta \right) = \frac{1}{5} \sin \theta \cos \theta$$

Using the double-angle identity $\sin(2\theta) = 2 \sin \theta \cos \theta$, we can rewrite the function as:

$$f(x) = \frac{1}{10} \sin(2\theta)$$

The sine function attains a maximum value of 1 (which occurs here when $2\theta = \frac{\pi}{2}$, or $\theta = \frac{\pi}{4}$, corresponding to $x = 5$). Thus, the maximum possible value of $f(x)$ is:

$$\frac{1}{10}$$

5. Evaluate the exact value of the product:

$$(\sin^4(75^\circ) - \cos^4(75^\circ)) \sec(150^\circ)$$

First, factor the difference of fourth powers inside the parentheses:

$$\sin^4(75^\circ) - \cos^4(75^\circ) = (\sin^2(75^\circ) - \cos^2(75^\circ)) (\sin^2(75^\circ) + \cos^2(75^\circ))$$

By the Pythagorean identity, the second factor is 1. The first factor resembles the double angle identity for cosine, $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$.

Factoring out a negative sign gives:

$$-(\cos^2(75^\circ) - \sin^2(75^\circ)) = -\cos(2 \times 75^\circ) = -\cos(150^\circ)$$

Now, substitute this back into the original product:

$$(-\cos(150^\circ)) \sec(150^\circ)$$

Because $\sec \theta$ and $\cos \theta$ are reciprocals ($\sec \theta = \frac{1}{\cos \theta}$), their product is 1.

Thus, the expression simplifies to:

$$-1$$

6. Given the equation $\frac{\sin x + \cos x}{\sin x - \cos x} = 3$, determine the exact value of $\sec^2 x$.

We can transform the given equation entirely in terms of tangent by dividing the numerator and denominator of the fraction by $\cos x$ (noting $\cos x \neq 0$ because if $\cos x = 0$, the ratio would be -1 , not 3):

$$\frac{\tan x + 1}{\tan x - 1} = 3$$

Now, solve this algebraic equation for $\tan x$:

$$\tan x + 1 = 3(\tan x - 1)$$

$$\tan x + 1 = 3 \tan x - 3$$

$$4 = 2 \tan x \implies \tan x = 2$$

To find $\sec^2 x$, we use the standard Pythagorean identity connecting tangent and secant:

$$\sec^2 x = \tan^2 x + 1$$

Substituting $\tan x = 2$ yields:

$$\sec^2 x = (2)^2 + 1 = 5$$